

Parameter-Efficient Adaptation of SAM 2 for Sentinel-1 SAR Flood Inundation Mapping in the Indo-Gangetic Region

Part 1: Motivation, Methods, Datasets

Aksan Gony Alif

Co-author: Mahbubur Rahman

BRAC University

Why this paper exists

- Bangladesh floods every monsoon. *Cloud cover* makes optical satellites unreliable when flood maps matter most.
- Sentinel-1 SAR **penetrates clouds**, available globally and free.
- But SAR is a noisy, single-channel, logarithmic-intensity signal — nothing like the natural RGB photographs vision foundation models (SAM, SAM 2) are pretrained on.
- **Question:** can we adapt SAM 2 to radar, *cheaply*, and have it generalize to an unseen flood event?

Answer (preview): yes, mostly — with one important caveat about the cross-polarization channel.

Three gaps in prior work

1. **No head-to-head comparison** of parameter-efficient fine-tuning (PEFT) methods on SAM-family backbones for SAR floods. Prior papers use one method each; you can't tell which adapter to pick.
2. **No test of operational utility** of uncertainty estimates. Prior SAR-flood work reports aggregate calibration (low ECE) but does not test whether the resulting *per-pixel* scores are useful for triage.
3. **No public reproducible OOD test set** combining temporal *and* geographic out-of-distribution shift on Sentinel-1. Standard benchmarks hold out one country, but all events are from 2017–2019.

We close all three.

Methodology: a controlled empirical sweep

5 backbones $\times$ 4 PEFT methods $\times$ 3 seeds

Backbone	LoRA	DoRA	Conv-LoRA	AdaptFormer
SAM ViT-B (94 M)	✓	✓	✓	✓
SAM 2 Hiera-B+ (81 M)	✓	✓	✓	✓
SAM ViT-L (308 M)			✓	
SAM ViT-H (636 M)			✓	
SAM 2 Hiera-L (224 M)			✓	

- Full sweep + baselines: $\approx$ **3.5 hours**, $\approx$ **\$10** cloud compute (dual NVIDIA RTX PRO 5000 Blackwell).
- Uncertainty: **Monte Carlo dropout**, 20 stochastic passes per chip.

Polarimetric “pseudo-RGB” input

SAM 2 was pretrained on three RGB channels. Sentinel-1 gives us only two:

- **VV** (co-polarization): vertical transmit, vertical receive.
- **VH** (cross-polarization): vertical transmit, horizontal receive.

We must synthesize a third channel.

Three compositions compared:

ratio	$(VV, VH, 10 \log_{10}(P_{VV}/P_{VH}))$
diff	$(VV, VH, VV_{dB} - VH_{dB})$
single	(VV, VV, VV)

By the dB-domain identity $10 \log_{10}(P_{VV}/P_{VH}) = VV_{dB} - VH_{dB}$, *ratio* and *diff* produce identical model inputs. The meaningful contrast is between dual-polarization (ratio or diff) and single-polarization (VV alone).

Four evaluation splits

Split	Chips	Role
Sen1Floods11 test	90	In-distribution (10 countries)
Sen1Floods11 Bolivia	15	Held-out country
Pakistan-S1F11 subset	12	Regional in-distribution (2010 event)
Pakistan-2022	39	Strict OOD: new geography + time

- The first three splits come from **Sen1Floods11** (Bonafilia et al., CVPR Workshops 2020). All events are from 2017–2019.
- The fourth, **Pakistan-2022**, is constructed in this work. It covers the August–September 2022 Indo-Gangetic monsoon flood — the same meteorological system that produces Bangladesh’s annual flooding.

Two public sources, paired in this work:

- **Labels:** TU Wien Sentinel-1-derived flood-extent product (Roth et al., NHESS 2023; CC-BY 4.0)
- **Imagery:** Microsoft Planetary Computer `sentinel1-1-rtc` (Radiometric Terrain Corrected, anonymous SAS-token access)

Pipeline: for each TU Wien mask date, find the matching Sentinel-1 RTC scene, read at native 10 m, reproject the mask upward to 10 m, tile into 512×512 chips.

Result: 39 chips at native 10 m, fully Sen1Floods11-compatible layout.

Acquisition note: the first chip set was at 20 m and mismatched the training distribution. We rebuilt at native 10 m; that is the dataset released here.

Training protocol:

- 10 epochs on the 252 Sen1Floods11 hand-labelled training chips.
- AdamW optimizer; separate learning rates 10^{-4} (adapters) / 10^{-5} (unfrozen pretrained weights); cosine schedule with 200-step linear warmup.
- Loss: Dice + binary cross-entropy with logits, equal weight.
- Mixed-precision fp16; batch size 1 with gradient accumulation 4 (effective batch 4).
- **No training-time data augmentation.** Test-time augmentation (4 rotations $\times$ 2 flips = 8 views) is applied only at inference for the ensemble condition.

Seeds: 42, 123, 20025 (three independent reruns per cell).

Setup complete.

The next half of this talk walks through what we found across these four splits — including one diagnostic finding that we did *not* expect:

The cross-polarization channel turns out to be the operative out-of-distribution failure mode.

Mahbubur Rahman will take it from here.